

SOFTWARE ENGINEERING

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We chose Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub System.

Reason 1: Fault Isolation for Retry/Dead Letter Handling (FR-001, FR-002)

It is important that any failed attempts be retried via an exponential backoff method and failed deliveries to be redirected to a Dead-Letter Queue without disturbing the primary ingestion pipeline. The microservices pattern will ensure fault isolation of the Retry and DLQ functions, thereby preventing downstream failure from affecting incoming webhooks.

Reason 2: Independence in scaling when loaded (NFR-001)

The ingestion endpoint should be able to handle up to 2,500 transactions per second with a reliability of 99.999%. Microservices provide independence in scaling for the Ingestion & Verification service compared to the Retry and Dead Letter Queue (DLQ) services which have relatively less and bursty traffic.

Security Advantage

Signature verification (FR-005, HMAC-SHA256) being an independent service implies that all unauthorized and/or forged requests are filtered out at the boundary level before reaching any of the Payload Delivery, Retry, or DLQ services, which confines the attack vector to the one component.

Performance Benefit

Payload Delivery, Retry, and Audit Logging are independent services interacting with each other via clearly defined interfaces; thus, even if the Target Service is lagging behind or unresponsive, it would not impede the Ingestion service from receiving new webhooks under the <50ms p95 latency requirement (NFR-002).